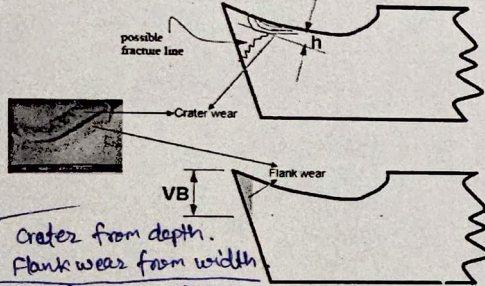


2

Tool Wear, Tool Life & Machinability

By S K Mondal

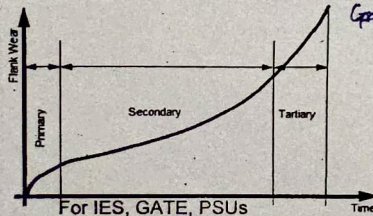
Tool Wear



Flank Wear: (Wear land)

Stages

- Flank Wear occurs in three stages of varying wear rates



For IES, GATE, PSU

Tool Failure

Tool failure is two types *Tool life, based on slow death*

- 1. Slow-death:** The gradual or progressive wearing away of rake face (crater wear) or flank (flank wear) of the cutting tool or both.
 - 2. Sudden-death:** Failures leading to premature end of the tool
- The sudden-death type of tool failure is difficult to predict. Tool failure mechanisms include plastic deformation, brittle fracture, fatigue fracture or edge chipping. However it is difficult to predict which of these processes will dominate and when tool failure will occur.

Flank Wear: (Wear land)

Reason

- Abrasion by hard particles and inclusions in the work piece.
- Shearing off the micro welds between tool and work material.
- Abrasion by fragments of built-up-edge ploughing against the clearance face of the tool.
- At low speed flank wear predominates.
- If MRR increased flank wear increased.

$MRR \uparrow \Rightarrow \text{flank wear} \uparrow$

Flank Wear: (Wear land)

Primary wear

The region where the sharp cutting edge is quickly broken down and a finite wear land is established.

Secondary wear

The region where the wear progresses at a uniform rate.

Crater wear increases exponentially with temp.

Tool Wear

- Flank Wear
- Crater Wear
- Chipping off of the cutting edge

Flank Wear: (Wear land)

Effect

- Flank wear directly affect the component dimensions produced.
- Flank wear is usually the most common determinant of tool life.

Flank Wear: (Wear land)

Tertiary wear

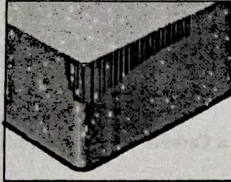
The region where wear progresses at a gradually increasing rate.

- In the tertiary region the wear of the cutting tool has become sensitive to increased tool temperature due to high wear land.
- Re-grinding is recommended before they enter this region.

Tool life criteria ISO

(A certain width of flank wear (VB) is the most common criterion)

- Uniform wear: 0.3 mm averaged over all past
- Localized wear: 0.5 mm on any individual past



Wear Mechanism

1. Abrasion wear *flank wear*
2. Adhesion wear *micro weld*
3. Diffusion wear *high temp. crater wear*
4. Chemical or oxidation wear *very-very high temp. + very high pr*
[Diamond tool flattened coolant is used? bcoz it reacts with O₂. Ferrus metal]
[Diamond is not used to cut steel]

Crater wear

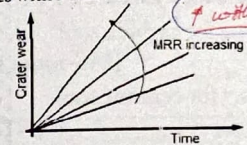
Due to diffusion of tool into chip in material

- More common in ductile materials which produce continuous chip.
- Crater wear occurs on the rake face. *at VF*
- At very high speed crater wear predominates
- For crater wear temperature is main culprit and tool diffuse into the chip material & tool temperature is maximum at some distance from the tool tip.

Crater wear

Contd.....

- Crater depth exhibits linear increase with time.
- It increases with MRR. *↑ with MRR ↑*

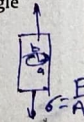


- Crater wear has little or no influence on cutting forces, work piece tolerance or surface finish.

Why chipping off or fine cracks developed at the cutting edge

[fine particles are removed from metal]

- Tool material is too brittle
- Weak design of tool, such as high positive rake angle *α↑*
- As a result of crack that is already in the tool
- Excessive static or shock loading of the tool.



stress concentration factor $K_t = 1 + \frac{2a}{b}$
Stress concentration is max at cutting edge

Notch Wear

Due to oxidation

- Notch wear on the trailing edge is to a great extent an oxidation wear mechanism occurring where the cutting edge leaves the machined workpiece material in the feed direction.
- But abrasion and adhesion wear in a combined effect can contribute to the formation of one or several notches.

Tool Life Criteria

Tool life criteria can be defined as a predetermined numerical value of any type of tool deterioration which can be measured.

some of the ways

- Actual cutting time to failure.
- Volume of metal removed.
- Number of parts produced.
- Cutting speed for a given time
- Length of work machined.

Taylor's Tool Life Equation

based on Flank Wear

Tool life is a fⁿ of V only.

Causes

- Sliding of the tool along the machined surface
- Temperature rise

$$VT^n = C$$

Where, V = cutting speed (m/min)

T = Time (min)

n = exponent depends on tool material only

C = constant based on tool and work material and cutting condition.

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Values of Exponent 'n'

$n = 0.08$ to 0.2 for HSS tool

$= 0.1$ to 0.15 for Cast Alloys

$= 0.2$ to 0.4 for carbide tool

[IAS-1999; IES-2006]

$= 0.5$ to 0.7 for ceramic tool

[NTPC-2003]

Extended or Modified Taylor's equation

$$VT^n f^a d^b = C$$

Where: d = depth of cut

f = feed rate

$$\text{or } T = \frac{C^{1/n}}{V^{1/n} f^{1/n_1} d^{1/n_2}}$$

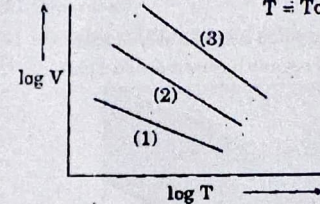
$$\frac{1}{n} > \frac{1}{n_1} > \frac{1}{n_2} \quad \text{effect } v > f > d$$

i.e Cutting speed has the greater effect followed by feed and depth of cut respectively.

Tool Life Curve

V = Cutting speed

T = Tool life



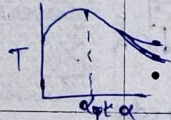
1. HSS

2. Carbide

3. Ceramic

Cutting speed used for different tool materials

HSS (min) 30 m/min < Cast alloy < Carbide
< Cemented carbide 150 m/min < Cermets
< Ceramics or sintered oxide (max) 600 m/min



Effect of Rake angle on tool life

- If rake angle is large, smaller will be cutting angle and larger will be the shear angle. This will reduce force and power of cut i.e. $\uparrow$ tool life.
- But increasing the rake angle reduces the mass of metal behind the cutting edge resulting in poor transfer of heat i.e. $\downarrow$ tool life.
- Therefore optimum value of $\alpha = 14^\circ$

$\alpha \uparrow \Rightarrow \text{Cut} \Rightarrow \phi \uparrow \Rightarrow \text{Force} \Rightarrow \text{Tool Life} \uparrow$

Effect of Clearance angle on tool life

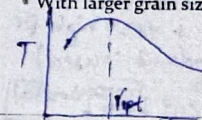
If clearance angle increased it reduces flank wear but weakens the cutting edge, so best compromise is 8° for HSS and 5° for carbide tool.

8° HSS

5° Carbide Tool

Effect of work piece on tool life

- With hard micro-constituents in the matrix gives poor tool life.
- With larger grain size tool life is better.

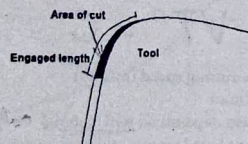


Grain size $\uparrow \Rightarrow$ Tool Life $\uparrow$

Chip Equivalent

$$\text{Chip Equivalent}(q) = \frac{\text{Engaged cutting edge length}}{\text{Plan area of cut}}$$

- It is used for controlling the tool temperature.



For IES, GATE, PSUs

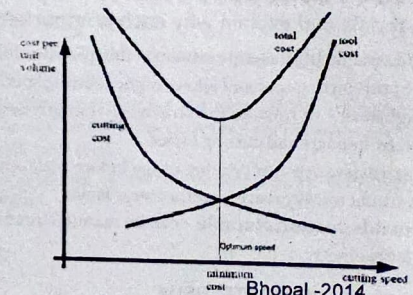
q more
Tool Life more

$G \uparrow \Rightarrow q \uparrow \Rightarrow T \uparrow$
 $r \uparrow \Rightarrow q \uparrow \Rightarrow \text{Tool Life}$

- The SCEA alters the length of the engaged cutting edge without affecting the area of cut. As a result, the chip equivalent changed. When the SCEA is increased, the chip equivalent is increased, without significantly changing the cutting forces.

- Increase in nose radius also increases the value of the chip equivalent and improve tool life.

Economics of metal cutting



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Formula

$$V_0 T_0^n = C$$

Optimum tool life for minimum cost

$$T_o = \left(T_c + \frac{C_1}{C_m} \right) \left(\frac{1-n}{n} \right) \quad \text{if } T_c, C_1, \text{ \& } C_m \text{ given}$$

$$= \frac{C_1}{C_m} \left(\frac{1-n}{n} \right) \quad \text{if } C_1 \text{ \& } C_m \text{ given}$$

Optimum tool life for Maximum Productivity
(minimum production time)

$$T_o = T_c \left(\frac{1-n}{n} \right)$$

Units: T_c - min (Tool changing time)

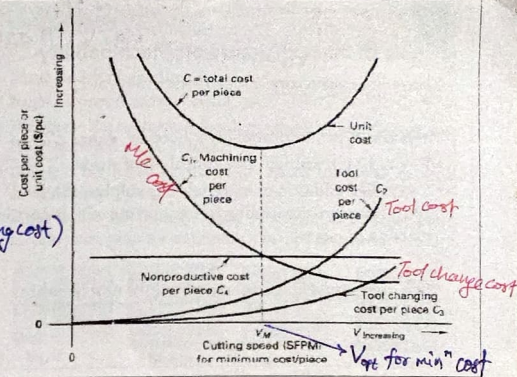
C_1 - Rs./ servicing or replacement (Tooling cost)

C_m - Rs./min (Machining cost)

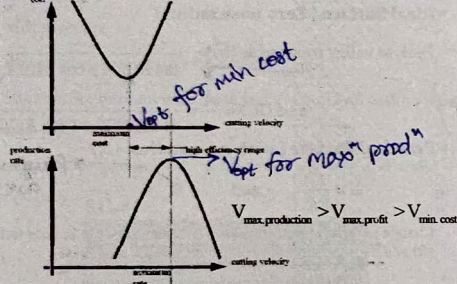
V - m/min (Cutting speed)

Tooling cost (C_1) = tool regrind cost (grinding time x machining cost) + tool depreciation per service/ replacement

Machining cost (C_m) = labour cost + over head cost per min



Minimum Cost Vs Production Rate



Machinability-Definition

Machinability can be tentatively defined as 'ability of being machined' and more reasonably as 'ease of machining'.

Such ease of machining or machining characters of any tool-work pair is to be judged by:

- Tool wear or tool life
- Magnitude of the cutting forces
- Surface finish *Copper & Brass excellent surface finish*
- Magnitude of cutting temperature
- Chip forms. *discontinuous chip (cast iron)*

Machinability Contd.....

- Machinability will be high when cutting forces, temperature, surfaces roughness and tool wear are less, tool life is long and chips are ideally uniform and short.
- The addition of sulphur, lead and tellurium to non-ferrous and steel improves machinability.
- Sulphur is added to steel only if there is sufficient manganese in it. Sulphur forms manganese sulphide which exists as an isolated phase and act as internal lubrication and chip breaker.
- If insufficient manganese is there a low melting iron sulphide will formed around the austenite grain boundary. Such steel is very weak and brittle.

For IES, GATE, PSUs

manganese sulphide acts as solid lubricant & chip breaker.

Free Cutting steels

- Addition of lead in low carbon re-sulphurised steels and also in aluminium, copper and their alloys help reduce their τ_c . The dispersed lead particles act as discontinuity and solid lubricants and thus improve machinability by reducing friction, cutting forces and temperature, tool wear and BUE formation.
- It contains less than 0.35% lead by weight.
- A free cutting steel contains C-0.07%, Si-0.03%, Mn-0.9%, P-0.04%, S-0.22%, Pb-0.15%

Machinability Index

Or Machinability Rating

The machinability index K_M is defined by

$$K_M = V_{60}/V_{60R}$$

Where V_{60} is the cutting speed for the target material that ensures tool life of 60 min, V_{60R} is the same for the reference material.

If $K_M > 1$, the machinability of the target material is better than this of the reference material, and vice versa

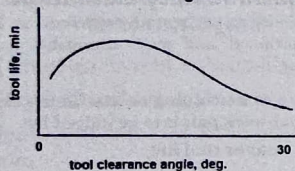
Role of microstructure on Machinability

Coarse microstructure leads to lesser value of τ_c .

Therefore, τ_c can be desirably reduced by Annealing $\Rightarrow \tau_{cs}$

- Proper heat treatment like annealing of steels
- Controlled addition of materials like sulphur (S), lead (Pb), Tellerium etc leading to free cutting of soft ductile metals and alloys.
- Brittle materials are relatively more machinable.

Effects of clearance angle on machinability



Inadequate clearance angle reduces tool life and surface finish by tool - work rubbing, and again too large clearance reduces the tool strength and tool life hence machinability.

Cutting fluid

- The cutting fluid acts primarily as a coolant and secondly as a lubricant, reducing the friction effects at the tool-chip interface and the work-blank regions.
- Cast Iron: Machined dry or compressed air, Soluble oil for high speed machining and grinding
- Brass: Machined dry or straight mineral oil with or without EPA.
- Aluminium: Machined dry or kerosene oil mixed with mineral oil or soluble oil
- Stainless steel and Heat resistant alloy: High performance soluble oil or neat oil with high concentration with chlorinated EP additive.

For IES, GATE, PSUs $\rightarrow$ Extreme pressure

Effects of tool rake angle(s) on machinability

$$\alpha_1 \Rightarrow m/c \uparrow$$

- As Rake angle increases machinability increases.
- But too much increase in rake weakens the cutting edge.

Effects of Nose Radius on machinability

Proper tool nose radiusing improves machinability to some extent through

- increase in tool life by increasing mechanical strength and reducing temperature at the tool tip
- reduction of surface roughness, h_{max}

$$h_{max} = \frac{f^2}{8R}$$

Effects of Cutting Edge angle(s) on machinability

- The variation in the cutting edge angles does not affect cutting force or specific energy requirement for cutting.
- Increase in SCEA and reduction in ECEA improves surface finish sizeably in continuous chip formation hence Machinability.

$$C_s \uparrow + C_e \downarrow \Rightarrow m/c \uparrow$$

Surface Roughness

- Idea Surface (Zero nose radius)

$$\text{Peak to valley roughness (h)} = \frac{f}{\tan SCEA + \cot ECEA}$$

$$\text{and } (R_s) = \frac{h}{4} = \frac{f}{4(\tan SCEA + \cot ECEA)}$$

- Practical Surface (with nose radius = R)

$$h = \frac{f^2}{8R} \quad \text{and} \quad R_s = \frac{f^2}{18\sqrt{3}R}$$

Change in feed (f) is more important than a change in nose radius (R) and depth of cut has no effect on surface roughness.

IAS -2009 Main

- What are extreme-pressure lubricants?

[3 - marks]

Where high pressures and rubbing action are encountered, hydrodynamic lubrication cannot be maintained; so Extreme Pressure (EP) additives must be added to the lubricant. EP lubrication is provided by a number of chemical components such as boron, phosphorus, sulfur, chlorine, or combination of these. The compounds are activated by the higher temperature resulting from extreme pressure. As the temperature rises, EP molecules become reactive and release derivatives such as iron chloride or iron sulfide and forms a solid protective coating.

